TrustVault - Comprehensive Test & Verification Report

Automated JUnit 5 Test Suite, Real MySQL Integration, and Live Runtime Verification

1. Test Execution Summary

A comprehensive test battery comprising automated unit tests, Spring Boot integration tests, real MySQL persistence verification, and live HTTP endpoint probing was executed.

All 32 automated tests passed with zero failures and zero errors on the active build.

- **Total Automated Tests:** 32 executed, 32 passed (100% pass rate).
- **Failures / Errors:** 0 Failures, 0 Errors, 0 Skipped.
- **Build Tool:** Apache Maven 3.9.11 with OpenJDK 22.0.2.
- **Live Server Verification:** Port 8082 started, verified 10 distinct UI and REST actions, and stopped cleanly.

2. Class-by-Class Automated Test Breakdown

Test Class	Tests Run	Passed	Failures	Errors	Execution Time
AuthControllerIntegrationTest	5	5	0	0	27.64 s
DocumentUploadSecurityTests	4	4	0	0	1.14 s
MySQLDatabaseCrudIntegrationTest	2	2	0	0	3.11 s
TrustVaultSecurityTests	21	21	0	0	12.99 s
TOTAL ACROSS ALL SUITES	32	32	0	0	44.88 s

3. Three Dedicated Test Accounts Verification

Account Role	Username	Provisioning Method	Login Status	Role Enforced
Dedicated Owner	tv_owner_test	Self-service registration (/api/auth/register)	VERIFIED (HTTP 200)	ROLE_OWNER
Dedicated Viewer	tv_viewer_test	Self-service registration (/api/auth/register)	VERIFIED (HTTP 200)	ROLE_VIEWER
Dedicated Admin	tv_admin_test	Authorized admin promotion (/api/admin/users/{id}/role	VERIFIED (HTTP 200)	ROLE_ADMIN

4. End-to-End Workflow & Security Evidence

- **Owner Upload Test:** Uploaded authentic PDF document (ID: 6, Title: 'Owner Confidential Financial Statement 2026'). Stored as UUID on disk, recorded in MySQL documents table, audit event logged.
- **Owner Download Test:** Owner successfully downloaded document (962 bytes). Content integrity matched 100%.
- **Viewer Upload Denial:** Viewer attempted multipart upload to /api/documents/upload. Backend returned HTTP 403 Forbidden.
- **Viewer IDOR Denial:** Viewer attempted unauthorized direct download of Owner's document. Backend returned HTTP 403 Forbidden.
- **Access Request Submission:** Viewer submitted access request (ID: 2) with justification 'Need compliance audit review'. Status set to PENDING.
- **Owner Request Approval:** Owner approved request 2 via /api/requests/2/review?approve=true. Status set to APPROVED and active READ permission created.
- **Watermarked Download:** Viewer downloaded authorized document (962 bytes). PDF was dynamically watermarked with viewer identity, timestamp, and IP.
- **Instant Revocation:** Owner revoked Viewer permission ID 5 via DELETE /api/permissions/documents/6/revoke/5. Access terminated immediately.
- **Post-Revocation Download:** Viewer immediately re-attempted download. Backend returned HTTP 403 Forbidden (Zero Grace Period verified).
- **Tamper Verification Check:** Admin executed /api/tamper/verify. Result: Valid: True, Broken Blocks: 0 across entire cryptographic chain.